



THEMIS User Manual

1. Introduction

Current Version

- Manual version: v0.2.2
- Compatible Minimum AOS version: v0.3.0
- Compatible Minimum AOS Server version: v0.3.0

1.1 Welcome and Overview

Welcome to the THEMIS Robot User Manual!

Congratulations on joining a growing community of researchers, educators, industrial professionals, and developers who are pioneering advanced robotics applications with THEMIS, the latest humanoid innovation from Westwood Robotics. THEMIS is a meticulously designed full-size, general-purpose humanoid robot, created with adaptability, strength, and precision in mind. It unlocks new possibilities across academic research, robotics education, human-robot interaction studies, complex service environments, and a range of industrial applications.

THEMIS stands at the forefront of humanoid robotics, integrating advanced technologies in proprioceptive actuation, mobility, perception, and control systems. Its robust hardware and sophisticated software architecture make it a versatile platform, capable of navigating and performing tasks in diverse and dynamic environments. Key features include high-performance

actuators, intuitive locomotion control, multi-sensor perception system, and an open platform that encourages customization and innovation.

In addition to its research and educational applications, THEMIS is engineered for industrial use, supporting operations in manufacturing, logistics, and other fields. Its human-like mobility and adaptability allow it to perform tasks in shared workspaces, material handling, assembly assistance, or even hazardous inspections that would otherwise put human workers at risk. With its open architecture, users in industrial settings can program and modify THEMIS to meet specific operational requirements, increasing efficiency and helping address workforce demands.

As an open platform, THEMIS provides users the flexibility to tailor functionalities according to specific research or industrial needs. Whether used for exploring human-like gait, testing new algorithms, enhancing workplace efficiency, teaching advanced robotics principles, or developing complex interaction scenarios, THEMIS offers extensive capabilities designed to support cutting-edge research, innovation, and industrial productivity. This manual will be your essential guide, enabling you to unlock the full potential of THEMIS and facilitating seamless integration into your projects.

1.2 Purpose of the Manual

The purpose of this manual is to provide you with comprehensive, step-by-step instructions for the safe and effective use of the THEMIS robot. It is designed to serve as a reference for new users, experienced operators, and technical personnel who need to gain an in-depth understanding of THEMIS's capabilities, operations, and maintenance. By following this manual, you will gain the knowledge required to fully harness the robot's potential while ensuring reliability and longevity.

This manual covers a range of essential topics:

- **Safety Guidelines:** Critical safety information to prevent accidents and ensure the safety for both the operator and the robot.
- **Setup and Configuration:** Detailed instructions on setting up THEMIS for the first time, including assembling components, calibrating sensors, and configuring software for specific tasks.
- **Hardware Overview:** An in-depth look at THEMIS's hardware, including specifications, actuator functions, sensor systems, and the modular components that support customization.
- **Software Interface and Controls:** Step-by-step guidance on the software platform, control interfaces, and programming environment, enabling you to operate THEMIS and implement custom functionalities.

- **Operational Procedures:** Guidelines on executing basic to advanced operations, including locomotion, object manipulation, perception tasks, and environment navigation.
- **Maintenance and Troubleshooting:** Best practices for routine maintenance, troubleshooting common issues, and ensuring optimal performance over the robot's operational lifespan.

Additionally, this manual includes tips on maximizing efficiency, ensuring safe use, and exploring the full scope of THEMIS's capabilities. Whether you are conducting rigorous testing in a laboratory or exploring educational applications in a classroom setting, this document is structured to support your specific needs, streamline your experience, and provide solutions to any challenges you might encounter.

With this manual, we aim to ensure you have all the resources needed to successfully operate, maintain, and develop innovative applications with THEMIS. Please refer to each section as required, and contact Westwood Robotics for any additional support.

1.3 Disclaimer and Warning

Please read this document thoroughly before using the product to ensure proper configuration and safe operation, as the information provided in this document is important for your safety, as well as your legal rights and responsibilities.

Failure to read and follow the instructions and warnings outlined in this document may result in serious injury to yourself or others, damage to the THEMIS Robot, or harm to nearby objects.

This document and any related materials, is subject to change at the sole discretion of Westwood Robotics. By using this product, you acknowledge that you have read and understood this disclaimer and warning, and agree that you are solely responsible for your actions while using the product, as well as any consequences that may arise.

You agree to use this product only for legitimate and lawful purposes, in full compliance with all applicable laws, rules, regulations, and any terms, precautions, practices, policies, or guidelines provided by Westwood Robotics Corporation.

Westwood Robotics Corporation accepts no liability for any damage, injury, or legal liability, whether direct or indirect, arising from the use of this product. The user is required to observe safe and lawful practices, including, but not limited to, those outlined in this document.

2. Safety Information

Ensuring the safe operation of the THEMIS humanoid robot is essential to prevent harm to users, bystanders, and the robot itself. Adhering to these safety guidelines will help maximize the performance, reliability, and longevity of THEMIS while minimizing risks.

2.1 Basic Safety Guidelines

Trained Operator Requirement: Only individuals who have completed the necessary training on the THEMIS robot's functions, control systems, and safety protocols should operate or maintain the robot. Untrained users may lack familiarity with emergency stop procedures and safe handling techniques, increasing the risk of accidents.

Safe Operating Environment: The robot should be used in a controlled, well-lit environment with ample space to accommodate its full range of motion. Avoid areas with excessive dust, debris, or unstable flooring that could obstruct or destabilize THEMIS during operation.

Dedicated Power Supply: Use only the batteries, power adapters and cables supplied or recommended by Westwood Robotics. Avoid using third-party chargers or adapters, as these may cause electrical faults, overheating, or component damage, which could lead to fire or electric shock.

Environmental Conditions: THEMIS is designed for indoor use and should not be exposed to water, extreme humidity, or temperatures outside the recommended operational range (e.g., 0°C - 40°C). Exposure to these conditions may damage internal components or reduce the robot's accuracy and reliability.

Emergency Stop Awareness: Ensure that all operators are familiar with the location and function of the emergency stop buttons and shutdown procedures. Having immediate access to stop controls can prevent accidents and protect personnel during unexpected movements or malfunctions.

2.2 Handling Precautions

Safe Lifting and Transportation: THEMIS is a sophisticated and delicate piece of equipment; handle it with care. Always use designated handles or grips, if available, when lifting or transporting the robot. Use a team of trained personnel or appropriate lifting equipment whenever the robot cannot be handled safely by one person. Avoid lifting from non-structural parts like joints, limbs, or sensors.

Battery Safety: Use only the official battery chargers and never leave THEMIS batteries unattended while charging. Keep the charging station clear of flammable materials and maintain a safe distance from the robot while charging. Never attempt to disassemble, puncture, or expose the battery to extreme temperatures, as this could lead to battery leaks, fire, or explosions.

Protective Equipment for Maintenance: Wear protective gloves, safety glasses, and other necessary protective equipment when performing maintenance, especially when working with actuators, joints, or electrical components. This reduces the risk of injury from unexpected movements, sharp edges, or electrical discharge.

Avoid Exceeding Load Capacity: Do not exceed the load capacity for THEMIS, as this can strain actuators and destabilize the robot. Carrying loads beyond recommended limits may lead to

excessive wear on the robot's joints, reduced battery efficiency, or component failure.

2.3 Operational Safety

Maintaining Safe Distances: During operation, maintain a safe distance from the robot and ensure all bystanders are aware of THEMIS' s range of motion. THEMIS' s arms, legs, and torso may move suddenly during programmed actions, and accidental contact could cause injury.

Clear the Operation Path: Before activating THEMIS, ensure its operating path is clear of objects, wires, or uneven flooring that could cause the robot to stumble or lose balance. THEMIS' s sensors help with navigation, but it' s best to provide an unobstructed path to minimize collision risk and improve operational efficiency.

Continuous Supervision: Never leave THEMIS unattended during active operation. A supervisor should be present to monitor its movements and take immediate action if needed, including halting the robot' s operation if it displays erratic or unsafe behavior.

Sensor and Camera Care: Do not cover or obstruct THEMIS' s sensors and cameras during operation. These components are essential for the robot' s spatial awareness and precise movements. Any obstruction can impair navigation, cause collisions, or disrupt its functionality.

Software and Firmware Updates: Regularly update THEMIS' s software and firmware to the latest version provided by Westwood Robotics. Updates often include security patches and functionality improvements essential for safe and reliable operation.

2.4 Emergency Procedures

Emergency Stop: If THEMIS displays erratic or unsafe behavior, immediately press the emergency stop button to halt all robot functions. Familiarize all operators with the emergency stop' s location and function, and ensure it is easily accessible at all times during operation.

Power Disconnection Protocol: If the emergency stop does not fully stop or disable THEMIS, disconnect the power source immediately to halt all functions. Only personnel trained in electrical and power-system safety should disconnect or handle the relevant power components.

Evacuation and Emergency Contact: If THEMIS shows signs of severe malfunction (e.g., smoke, unusual sounds, or overheating), evacuate the immediate area and contact the necessary emergency personnel. Avoid approaching the robot in such cases, as it may present electrical or mechanical risks.

Battery Emergency Response: If the battery shows signs of malfunction, such as swelling, leaking, or overheating, do not touch, disassemble, or attempt to remove the battery. Instead, ventilate the area, avoid inhaling any fumes, and allow the battery to cool naturally. Contact Westwood Robotics support or trained personnel for safe handling and disposal.

Incident Documentation and Reporting: After any emergency situation, thoroughly document the incident, including the time, observed issues, actions taken, and any injuries or damages.

Submit a copy of the incident report to Westwood Robotics Support to initiate further assistance, repairs, or safety analysis.

Inspection After Emergency: Following any emergency or unexpected event, inspect THEMIS for signs of damage. Do not operate the robot again until it has been cleared by a certified technician or other authorized personnel. Unchecked mechanical or electrical issues may lead to further hazards or equipment failure.

3. Getting Started

THEMIS is carefully packed into two identical, robust shipping boxes designed to protect each part of the robot during transit. Before proceeding, read through the entire unboxing guide to ensure that the robot is unpacked safely and completely.

3.1 Unboxing

3.1.1 What's in the Boxes

THEMIS is shipped in two protective cases, one containing the Upper Body and the other the Lower Body. Inside these cases, you will find everything you need to begin setup. Each item in the boxes is essential to the initial setup and safe operation of THEMIS.

THEMIS Upper Body Case

- Upper Body: This includes the torso, arms, and head components.
- Batteries (x2): These are essential for powering THEMIS and are packaged separately within the box. Only use these provided batteries for optimal performance.
- Calibration Tools & Bolts: Required for precise alignment of the robot's joints during calibration. Bolts of the specified sizes are provided to secure the calibration tools.
- Assembly Bolts: Bolts specifically for connecting the upper and lower body securely.



Important Safety and Handling Precautions

THE TOTAL WEIGHT OF THE THEMIS LOWER BODY CASE EXCEEDS 30 KG. PLEASE LIFT CAREFULLY AND CONSIDER ASSISTANCE IF NEEDED.

- Always handle the components with care, as they contain delicate sensors and electronic parts.
- Be mindful when unpacking to avoid accidental drops or impacts, especially the batteries and calibration tools.
- Ensure all unpacking occurs on a clean, flat surface to reduce the risk of losing small components or bolts.

3.1.2 Unpacking Instructions

Follow the instructions below to unpack your THEMIS and prepare it for assembly.

THEMIS Upper Body

1. Unlock the 5 safety locks (one on each side and three on the front) to open the THEMIS Upper Body Case.
2. Temporarily remove the Calibration Tools & Bolts and the Assembly Bolts from the box to access the first layer of protective foam. Place these items in a secure location or return them to the case at the end for safekeeping.
3. Lift and remove the first layer of protective foam to access the batteries. Carefully remove the batteries from the case.
4. Gently lift the upper body of THEMIS from the case and lay it on a secure, clean, flat surface, such as a benchtop or the floor.
5. Return the protective foam layers to the case.
6. Close the lid and relock the safety locks to secure the case.

THEMIS Lower Body



HEAVY ITEM

The THEMIS Lower Body weighs over 20 kg. Lift with care, and seek assistance if needed. We recommend wearing protective gloves for additional protection.

1. Unlock the five safety locks (one on each side and three on the front) to open the THEMIS Lower Body Case.
2. Remove the first layer of protective foam to reveal the left leg.
3. Carefully lift the left leg and remove the second layer of protective foam, including any foam blocks between the left and right leg pitch actuators.
4. Gently lift the lower body of THEMIS from the case, then place it on a sturdy chair with a hard, flat seat or lay it on a secure, clean, flat surface, such as a benchtop or the floor.
5. Return the protective foam layers to the case.
6. Close the lid and relock the safety locks to secure the case.

3.1.3 Assembly Instructions

To assemble THEMIS, follow the steps below. For safety and ease of assembly, it is highly recommended to have at least two people involved in the process. Ideally, a gantry crane should be used to lift the upper body, as it can reduce the risk of damage and ensure smooth alignment.

1. **Prepare the Assembly Area:** Place the Lower Body on a flat, hard-surfaced chair or the surface of the protective case, ensuring the robot is stable and secure.

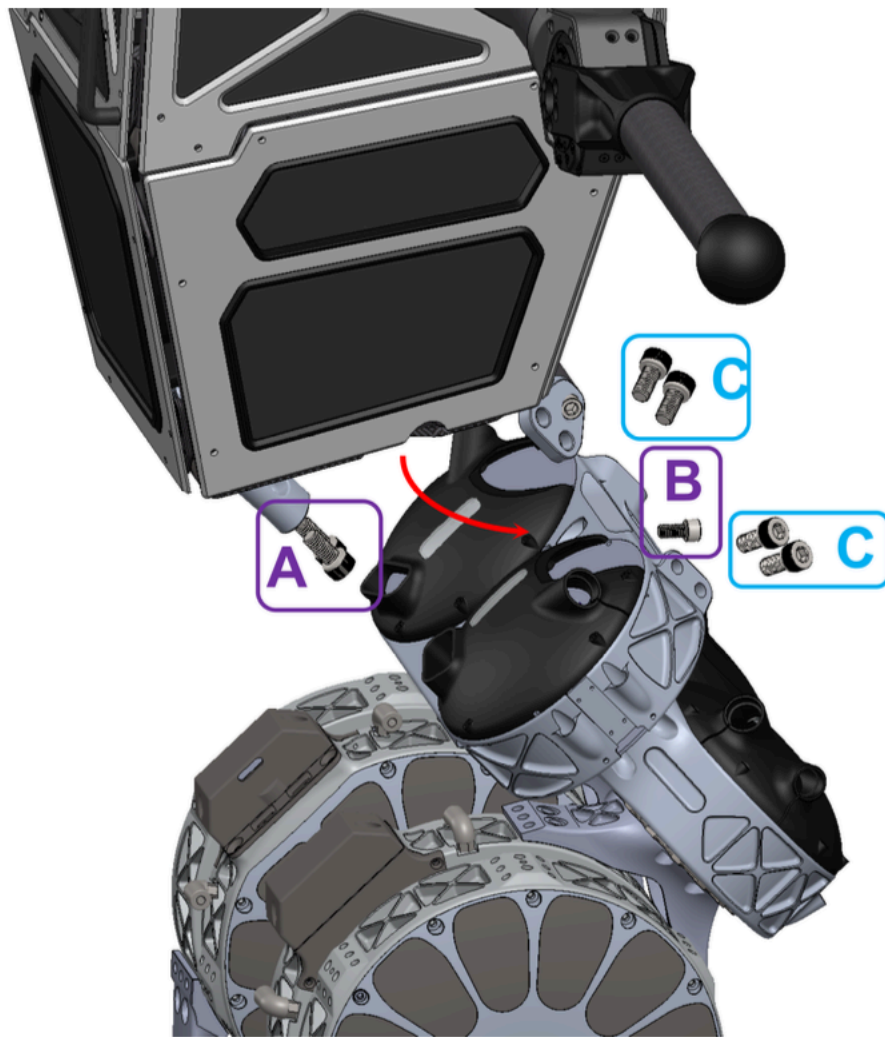


Figure 3.1: Assembling the Upper Body and the Lower Body

2. **Align the Upper and Lower Bodies:** Carefully lift the Upper Body and align it with the Lower Body, as shown with the **red arrow** in [Figure 3.1](#). Ensure both parts are positioned correctly before proceeding with the connection.



TEAMWORK

Aligning the parts may require two people for stability and safety.

3. **Secure the Upper Body to the Lower Body:** Once the upper and lower bodies are aligned, use the Assembly Bolts to secure the two parts together, following the steps below. Referring to [Figure 3.1](#) for bolt locations):
 - Secure the **Upper Body** onto the **Lower Body** with **A - M8 x 16 Hex-socket Screw x 2** and **B - M6 x 12 Hex-socket Screw x 1**.

- Secure the back of the **Upper Body** onto the **Lower Body** with **C - M8 x 16 Hex-socket Screw x 4**.
- Tighten the bolts evenly, ensuring a firm connection.

AVOID DAMAGE

Pay close attention to any **cables** between the upper and lower bodies to avoid pinching or damaging them during assembly. Take extra care when moving the upper body into place.

4. **Connect the Cables:** At the front of the robot, between the upper and lower bodies, connect the necessary cables for each leg:
- **Signal Cables:** Connect the signal cable to each leg first.
 - **XT60 Power:** Connect the XT60 power connectors to each leg.
 - **XT90 Power:** Connect the XT90 power connectors to each leg.

Handling

Ensure that all connections are secure, and avoid forcing any connectors into place.

5. **Final Check:** After securing the upper body, double-check all connections and bolts to ensure everything is tight and properly aligned before powering on the robot.

3.2 THEMIS Overview

3.2.1 Introduction to THEMIS

THEMIS is a full-size humanoid robot designed to excel across diverse domains, from advancing research and education to meeting the demands of commercial applications. Developed by Westwood Robotics, THEMIS embodies cutting-edge engineering and human-like versatility, making it an ideal platform for innovation and real-world problem-solving.

Standing at an adult human height, THEMIS is equipped with advanced locomotion, precise manipulation capabilities, and a robust sensory suite. Its modular architecture and intuitive software allow it to adapt seamlessly to various use cases, including industrial automation, customer interaction, and beyond.

Designed for scalability and efficiency, THEMIS bridges the gap between cutting-edge technology and practical applications. Whether you're exploring AI and robotics research,

enhancing educational initiatives, or deploying in commercial settings, THEMIS provides an ideal balance of precision and reliability.

Transform the future of robotics with THEMIS — a versatile platform for limitless possibilities.

3.2.2 Component Overview

The THEMIS robot is equipped with various advanced components designed to provide optimal performance, precise control, and enhanced sensing capabilities.

Key Components

- **Head Assembly:** At the top of the robot, THEMIS is equipped with a ZED 2i stereo camera paired with a Jetson Orin unit for depth perception and visual processing, allowing the robot to perceive its surroundings in real-time. A speaker and a microphone are also included in the head system and connected to the Jetson Orin unit. The head suite includes a 2-DoF neck driven by Koala BEAR proprioceptive actuators for swift and precise motion.
- **Upper Body and Torso:** The upper body houses the main control computer, which is responsible for high-level processing and coordinating the robot's operations, such as manipulation and locomotion.
- **Arm Modules:** THEMIS's 4-DoF arms are equipped with Koala BEAR Muscle Build proprioceptive actuators for manipulation tasks. The actuators provide dexterous force-controlled motion, suitable for research and practical applications.
- **Lower Body and Leg Actuators:** The lower body of THEMIS contains powerful actuators designed to provide the necessary force for dynamic locomotion. The actuators are arranged to allow for human-like walking and balancing.
- **Foot Modules with Contact Sensors (Optional):** THEMIS can be equipped with specially designed foot modules that incorporate foot contact sensors, which are 1-DoF force sensors that are designed to withstand high impact loads. Customers can choose to mount up to three of these sensors on the three "corners" of the foot to provide feedback on ground contact to the robot's control system.
- **Liquid Cooling System (Optional):** The torso can optionally be equipped with a liquid-cooling system to prevent overheating of the actuators, ensuring stable operation during intensive tasks.
- **Battery Compartment:** Integrated into the torso is the battery compartment, which houses two 8S Li-Ion batteries that power all of THEMIS's systems. The batteries are easily accessible for maintenance and replacement.

Component Overview Diagram

Figure 3.2 below highlights the main components of THEMIS and their functions. It provides a visual breakdown of the main components, allowing users to familiarize themselves with the

physical structure and key components of THEMIS.

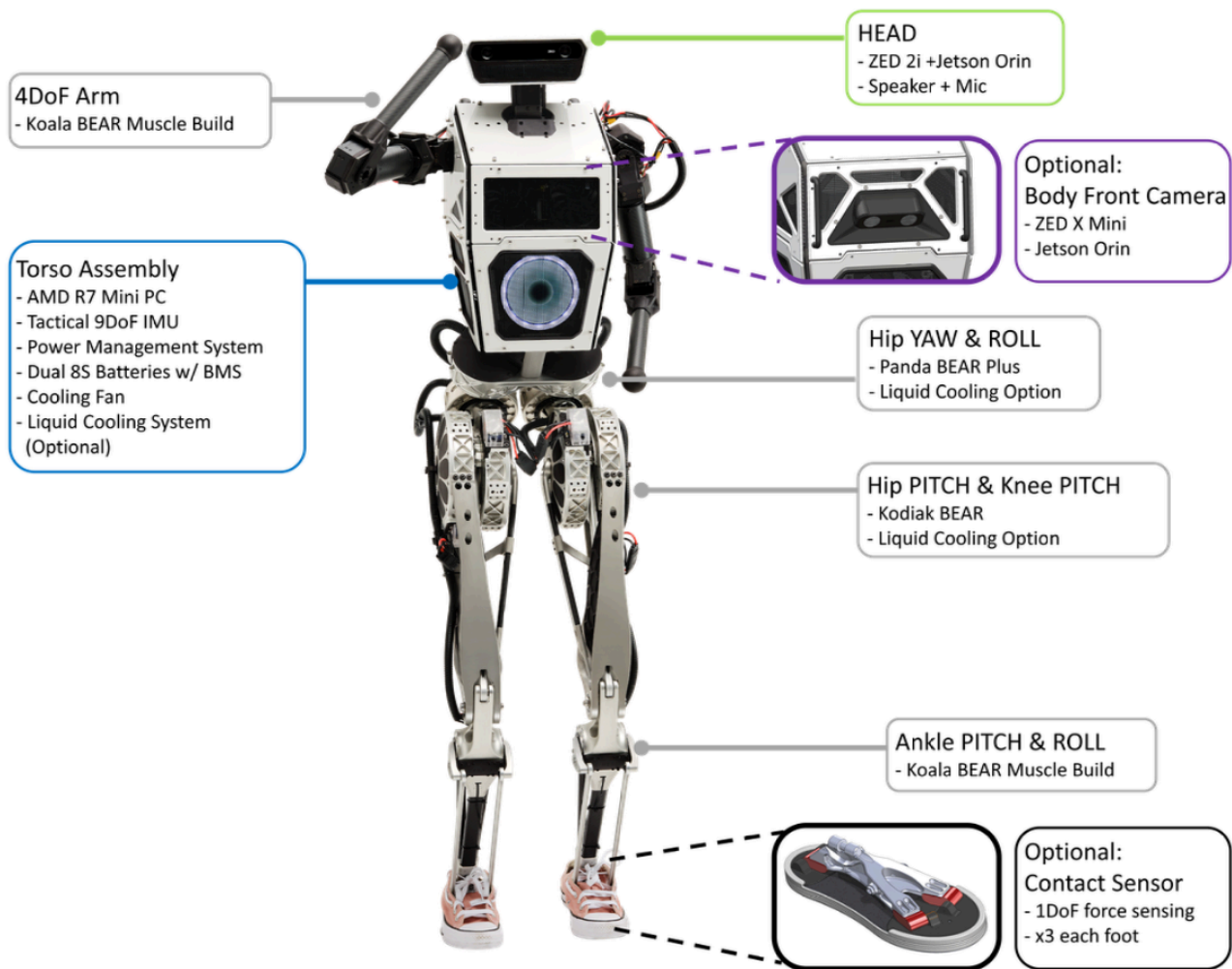


Figure 3.2: Component Overview of THEMIS

3.2.3 Power System Diagram

The power system of THEMIS is designed to efficiently manage and distribute energy to various components of the robot. It is powered by two 8S Li-Ion batteries, which serve as the primary power source. The batteries are connected to the Power Management System (PMS), responsible for monitoring, regulating, and distributing power across the robot's subsystems.

Key Components of the Power System

- **Batteries:** Two 8S Li-Ion batteries serve as the primary energy source for THEMIS.
- **Power Management System (PMS):** Contained in the Power Hub, the PMS ensures safe and efficient distribution of power from the batteries. It includes a built-in DC-DC converter for generating specific voltage levels required by various components.
- **Dedicated DC-DC Converter:** This converter provides a stable 19V power supply to the onboard computers.

Power Distribution Overview

1. **Direct Power Supply:** The batteries directly supply power to the actuators, ensuring optimal energy delivery for performance-critical operations.
2. **Onboard Computers:** A dedicated DC-DC converter is used to regulate power from the batteries and provide 19 V for the onboard computing systems.
3. **PMS-Managed Power Supply:** The PMS' s internal DC-DC converter generates power for the following:
 - **12 V Output:** Powers the cooling system and other general-purpose components.
 - **5 V and 3.3 V Outputs:** Provide power to the PMS internal logic, the built-in cooling system controller, the wireless emergency stop (E-stop) system, and foot-contact sensors.

Power-System Diagram

Refer to [Figure 3.3](#) for a visual representation of the power system. The diagram illustrates the connections between the batteries, PMS, actuators, onboard computers, and other components.

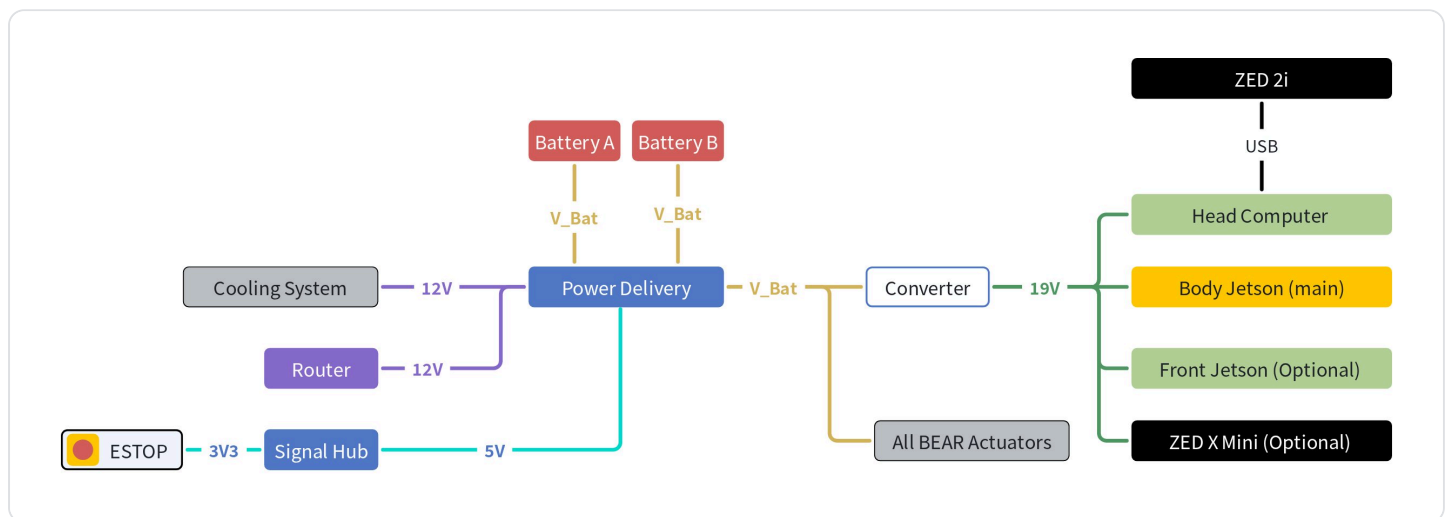


Figure 3.3: Power Distribution Diagram of THEMIS

3.2.4 Signal Diagram

The signal system of THEMIS is designed to facilitate efficient communication between its key components, ensuring seamless operation and precise control. The diagram below illustrates the connections between the robot's sensors, computers, and actuators, as well as their relationship to the power hub.

Key Components and Connections

- **BEAR (Black Boxes in Diagram):** The actuators responsible for motion control are connected through designated signal chains for the right leg, left leg, and upper body.
- **E-STOP System (Red Lines):** The wireless emergency stop (E-STOP) signals are transmitted to the Main PC to ensure immediate operational halt when required.
- **Foot Contact Sensors (Gray Boxes):** These optional sensors provide real-time ground-contact feedback to augment the robot's state-estimation system.

- **Computers:**
 - **Head Computer:** Jetson Orin integrated with the ZED 2i camera for multi-purpose depth sensing. Connected to other computers via Ethernet.
 - **Front Computer (Optional):** A Jetson Orin integrated with the ZED X Mini camera for detailed perception of the near field and the surrounding ground area. Connected to other computers via Ethernet.
 - **Main Computer:** Central control unit running controllers for locomotion and navigation, managing signals from the actuators, sensors, and other computers.
- **Power Hub (Blue Box):** The **Power Hub** contains the **Power Management System (PMS)** and serves as the central hub for signal and power distribution.

Signal Flow Description

1. Actuator signals are divided into three primary chains:
 - **Right Leg Chain (Yellow Lines):** Transmits control and feedback signals for the actuators in the right leg.
 - **Left Leg Chain (Orange Lines):** Transmits control and feedback signals for the actuators in the left leg.
 - **Upper Body Chain (Brown Lines):** Handles signal communication for the actuators in the upper body.
2. The emergency stop signal (red lines) is routed directly to the actuators via the Power Hub to ensure a rapid response during critical situations.
3. Foot contact sensors (gray boxes and dashed lines) provide stability and posture-related feedback to the computers.
4. The Ethernet connections (purple lines) establish communication between the computers, including optional connections to external networks via a router.
5. The Power Hub connects to the relevant components and supports centralized signal and power distribution.

Optional Features

Advanced versions of THEMIS may include optional features, such as additional Ethernet connections to optional computing units or sensors, as indicated by the dashed purple lines. These features extend the robot's capabilities for more complex applications.

Signal Diagram Visualization

Figure 3.4 provides an overview of the signal paths that enable THEMIS to function efficiently and respond to environmental feedback.

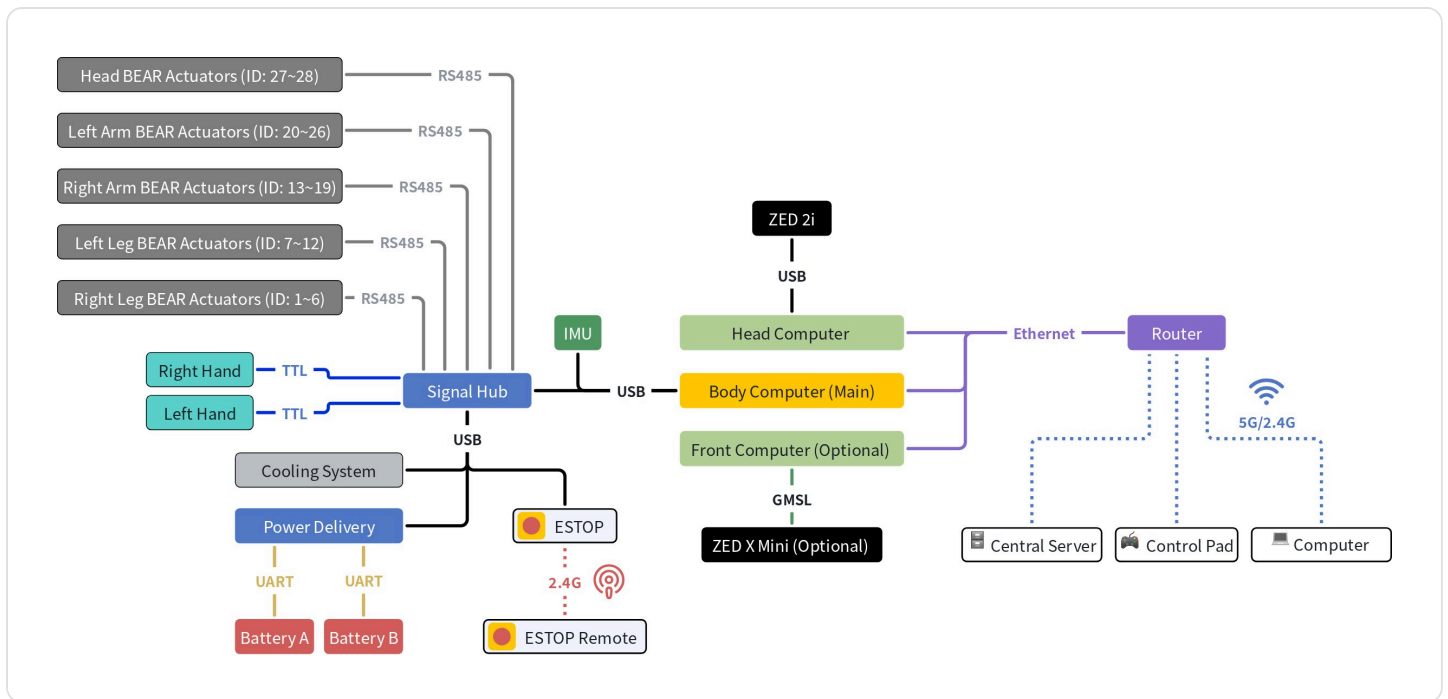


Figure 3.4: Signal Diagram of THEMIS

3.3 Operation

3.3.1 Powering On/Off

Power On: To power on the THEMIS robot, complete the following steps:

1. Wake the Batteries (refer to the [Batteries](#) section for more details on battery operations):
 - (a) Press the WAKE button on each of the two THEMIS batteries to activate them.
 - (b) Confirm that both batteries have entered Normal mode and are ready for operation.
2. Check the Battery Level:
 - (a) Press the BAT button once on each battery to display its charge level.
 - (b) Verify that both batteries are at least 50% charged to ensure sufficient operating time and proper performance.
3. Install the Batteries:
 - (a) Carefully slide the two battery slots in the battery compartment on the robot.



Dual-Battery Operation Required

Install both batteries for operation. The robot will be unstable and actuators will report faults if only one battery is installed during operation.

(b) Push gently until each battery clicks into place, ensuring they are firmly and securely installed.

(c) Close and lock the battery compartment lid before proceeding.



Secure & Lock

Make sure both batteries are firmly and securely installed. Lock the lid before operation.

4. System Boot-Up:

(a) Once the batteries are installed and the compartment is secured, the robot's computers will begin their startup process automatically.

(b) The Main PC typically boots up faster than the Jetson Orin units, and it takes about 50 s for the router to fully boot up.

(c) As soon as the computers and the router are ready, you may connect to the robot using the Controller UI or via SSH.

Power Off: To power off the THEMIS robot, follow these steps:

1. **Initiate Shutdown:** Use the gamepad's designated shutdown command (refer to Section 3.3.3), or run the power-off bash script from a terminal session connected to the main computer: `Main PC/THEMIS/THEMIS/Util/poweroff.sh`.
2. **Wait for Shutdown Completion:** Monitor the robot's status indicators and ensure that all onboard computers and systems have fully powered down. Do not remove the batteries until shutdown is complete to prevent data loss or hardware damage.
3. **Remove the Batteries:** Once all computers and systems are off, open the battery compartment and carefully remove the batteries. Store the batteries according to the recommended guidelines to maintain battery health and extend service life.



Avoid Damage to the Computers

Always follow these steps in order. Removing the batteries before the computers have fully powered

down may result in corrupted data or reduced component lifespan.

3.3.2 Battery Replacement

When to Change the Batteries: It is recommended to replace the batteries when the voltage drops below 23 V to ensure optimal performance and prevent unexpected shutdowns.

Preparing the Replacement Battery:

1. Wake the Battery: Press the WAKE button on the new battery to activate it.
2. Check Battery Level: Press the BAT button once and verify that the battery is at least 50% charged before installation.

Battery Replacement Options:

A. Full Power-Off Replacement:

- a. Shut Down the Robot: Follow the Power Off procedure in Section 3.3.1 to fully power down THEMIS.
- b. Remove the Old Battery: Open the battery compartment, carefully remove the old battery, and set it aside.
- c. Install the New Battery: Slide a replacement battery that has been awakened and has sufficient charge into the compartment until it is firmly seated.
- d. Close and Lock the Compartment: Secure the battery compartment lid before powering the robot back on.

B. Hot Swapping (Beta Feature):



Beta Feature

Hot swapping is currently in beta and may not be as stable as a full power-off replacement.

(a) Prepare the Robot:

- i. Move THEMIS onto a flat, stable surface.
- ii. Ensure the robot is in a stable stance mode with both the body-orientation offset and the center-of-mass (CoM) offset set to zero.
- iii. Cease all operations and movements before proceeding.
- iv. Secure the robot using a suitable gantry hoist.



Secure the Robot

Hot swapping is currently in beta. Secure the robot to a gantry to avoid damage in case of an unexpected loss of power.

(b) Swap the Batteries one by one:

- i. Open the battery compartment.
- ii. Remove and replace batteries individually to maintain uninterrupted power to the robot.

iii. Do not operate the robot or perform any other tasks during this process.

(c) Close and Lock the Compartment: Once the new batteries are installed, securely close and lock the battery compartment.

After completing either method, the robot should be ready to continue operation, provided that the batteries

are sufficiently charged and the compartment is properly secured.

3.3.3 Controller UI

Refer to [AOS Controller Manual](#) for details on how to control THEMIS via the AOS Controller UI.

4. Robot System

4.1 Actuators

All actuators on THEMIS are BEAR series proprioceptive actuators produced by Westwood Robotics. Despite customized housing and mounting features, the performance of these actuators on THEMIS is identical to the standard BEAR series actuator modules, unless otherwise stated.

The actuator types and specifications are listed in Table 4.1. For technical guidance and SDK usage, refer to the [SDK Manual of BEAR Actuators](#).

Property of BEAR	Koala BEAR	Koala BEAR Muscle Build	Panda BEAR	Panda BEAR Plus	Kodiak BEAR
Current Version	KB02	KBMB01	PB02	PB02P	CB01
Applied Joints	Neck	Arm / Ankle	N.A. N.A.	Hip Yaw / Hip Roll	Hip Pitch / Knee Pitch
Weight	250 g	285 g	685 g	925 g	2500 g
Supply Voltage	9~33.6 V (3~8 S)		9~50.4 V (3~12 S)		
Power Connector	XT30		XT60		XT90
Signal Connector	Molex PicoBlade 53047 6Pin				

Reflected Inertia, (kg/m ²)	1.82×10^{-3}		7.44×10^{-3}		0.1153
Speed Constant KV	27.3 RPM/V	9 RPM/V	14.3 RPM/V	7.1 RPM/V	3.3 RPM/V
Torque Constant KT	0.35 Nm/A	1.16 Nm/A	0.67 Nm/A	1.3 Nm/A	2.86 Nm/A
Stall Torque 15sec	3.5 Nm	8 Nm	13.4 Nm	26.5 Nm	180 Nm
Stall Torque 15sec (LC)	4.2 Nm	N.A. ²	16.8 Nm	33 Nm	240 Nm
Stall Torque 1.5sec	10.5 Nm	20 Nm	33.5 Nm	67 Nm	≥ 350 Nm

Table 4.1: Specification of BEAR on THEMIS¹

1 Stall torque is the maximum torque a BEAR actuator can deliver for the specified duration while limiting the temperature rise to no more than 55°C.; LC stands for Liquid Cooling.

2 Liquid cooling option is not available on KBMB01.

4.2 Sensors

4.2.1 Inertial Measurement Unit (IMU)

The main Inertial Measurement Unit (IMU) used for motion control is the HBK Microstrain [3DM-CV7-AHRS](#). It is capable of providing linear acceleration, angular rate, and Euler angles at a frequency of 1 kHz via the [Microstrain's official SDK](#). The device is located in the chest, in front of the main computer.

4.2.2 Cameras

THEMIS comes with a [ZED 2i](#) stereo camera on the head and a [ZED X](#) Mini stereo camera mounted on the front of the torso (on the advanced version).

- The **ZED 2i** on the head is responsible for primary visual perception and is used for general object recognition, target tracking, and visual interaction. In addition, it has built-in sensors that add localization and motion-sensing capabilities to applications, including an accelerometer, gyroscope, temperature sensor, barometer, and magnetometer. These sensors can be used to detect camera movements, determine camera orientation relative to magnetic north, detect relative altitude variations, analyze external weather conditions, and so on.
- The **ZED X Mini** on the front chest is responsible for auxiliary obstacle perception in navigation and planning, as well as compensate for the head camera's blind spots.

Detailed information is listed in Table 4.2.

Table 4.2: Camera Specifications

	ZED 2i	ZED X Mini
Sensor Type	1/3” 4MP CMOS	1/2.6” 2.3MP RGB
Array Size	2688 x 1520 pixels	1928 x 1208 pixels
Pixel Size	2µm x 2µm	3µm x 3µm
Shutter	Electronically synchronized rolling shutter	Electronically synchronized global shutter
Output Format	YUV 4:2:2 - UYV (8 bits)	RAW10
Max S/N Ratio	38.3 dB	38 dB
Dynamic Range	64.6 dB	71.4 dB
Sensitivity	1900 mV/Lux-sec	22.3 Ke/Lux-sec

4.3 Power Management

4.3.1 Battery

Specifications: THEMIS uses two 8 S lithium ion rechargeable batteries, and the specifications of the batteries are listed in Table 4.3.

Operation: As shown in Figure 4.1, there are two buttons and an array of LEDs located on the front of the battery. The LEDs are used to display battery status, and the buttons are used to interact with the battery.

The battery has four different modes:

- **Normal:** This is the battery's default operating mode after startup. Press the **BAT** button once while in **Normal** mode to show current battery level via the LEDs. The user should always place the battery into **Normal** mode before inserting the battery into the robot.

Property	Value	Note
Rated Capacity	9000 mAh	
Nominal Voltage	28.8 V	
Max. Charge Voltage	33.2 V	
Discharge cut-off voltage	20 V	Enforced by BMS
Recommended Charge Current	4.5 A	
Max. Charge Current	15 A	Reduced service life
Discharge Current	45 A	Continuous
	90 A	Short-term
	200 A	Peak, enforced by BMS
Operating Temperature	0 to 45 °C	Charge
	-20 to 60 °C	Discharge
Storage Temperature	-10 to 25 °C	
Weight		
Dimension		

Table 4.3: THEMIS Battery Specifications

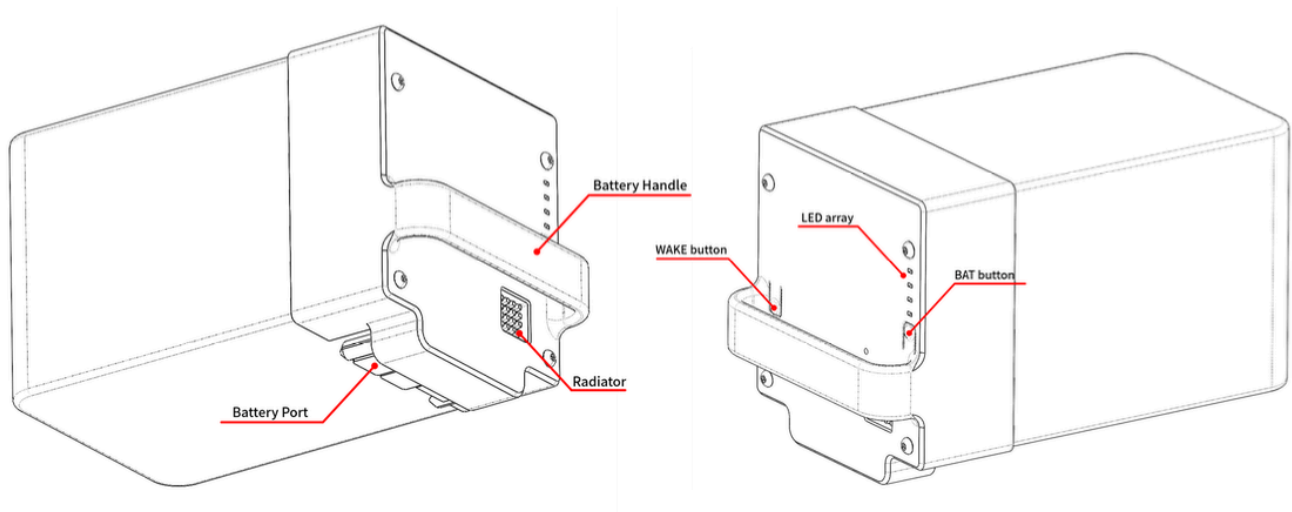


Figure 4.1: THEMIS Battery

- **Sleep:** The battery cannot be charged or discharged in **Sleep** mode, and the BMS monitors the battery at a significantly reduced frequency to preserve energy. Single click the BAT button while in **Sleep** mode to show present battery level via the LEDs. The battery can only be put into **Sleep** mode from **Normal** mode.

To put the battery into Sleep mode: while the battery is in **Normal** mode, single click then press and hold the **BAT** button, wait until the LEDs light up in sequence then turn off, then the battery enters **Sleep** mode. If the button is released before the LEDs all light up, the operation will be canceled and the battery remains in **Normal** mode.

To Exit Sleep mode: while the battery is in Sleep mode, single click then press and hold the **BAT** button, wait until the LEDs light up in sequence and remain lit, then the battery enters Normal mode. If the button is released before the LEDs all light up, the operation will be canceled and the battery remains in **Sleep** mode.

- **Shutdown:** The battery can not be charged or discharged in **Shutdown** mode, and the BMS does not monitor the battery, and the charge level cannot be displayed. This mode is intended for battery storage.

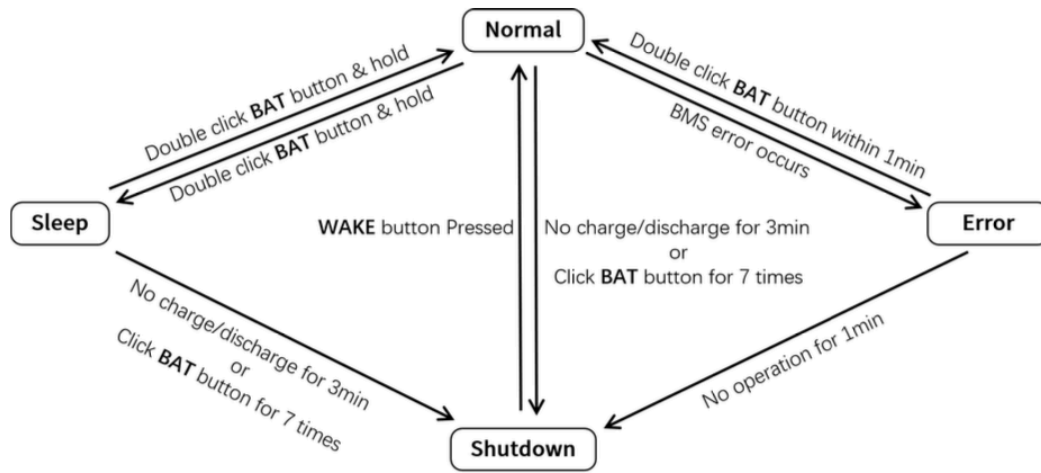


Figure 4.2: Modes of THEMIS battery

Entering **Shutdown** mode: The battery will enter **Shutdown** mode automatically if it is not being charged or not in use for more than 3 minutes. Users can also manually put battery into **Shutdown** mode from **Normal** mode or **Sleep** mode by clicking the **BAT** button more than 7 times within 5 seconds.

Waking up from **Shutdown** mode: The battery can only be woken up from **Shutdown** mode manually. While

The battery is in **Shutdown** mode, press the **WAKE** button to wake it up into **Normal** mode. Upon wake-up, all LEDs will light up then go off.

- **Error:** This mode will be triggered when a BMS error occurs while the battery is in **Normal** mode, and the battery can not be charged or discharged in **Error** mode. For the first 1 minute of Error mode, the error code will be displayed by blinking the LED array, and user can double click the **BAT** button to reset the battery. If no operation for more than 1 minute, the battery will enter **Shutdown** mode automatically.

Charging: Before charging, connect the battery adapter to the charger as shown in Figure 4.3, make sure that both the balance lead and the power cable are securely connected, then connect the charger to the power supply. To power up the charging system, connect the power supply to a compatible 110V/220 V AC power outlet.

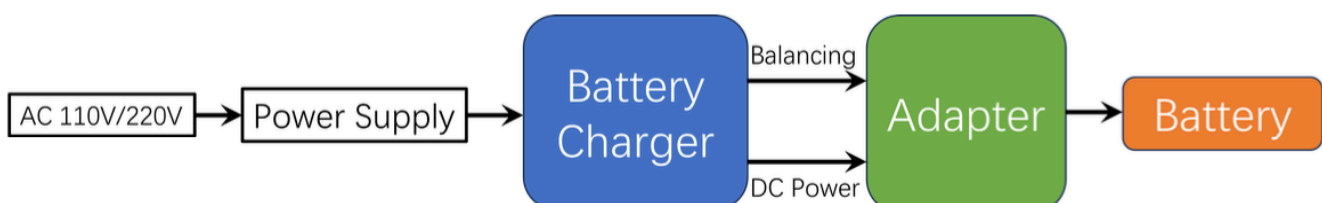
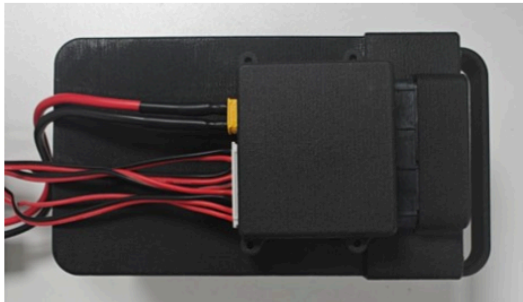
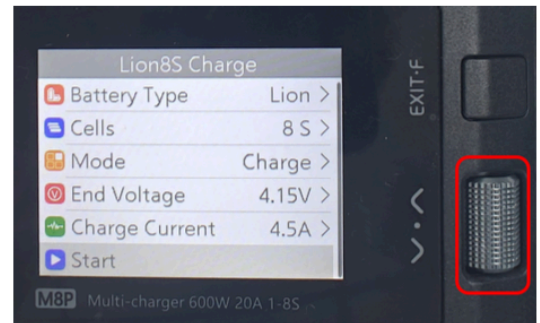


Figure 4.3: THEMIS Battery Charging Setup

Make sure to wake the battery and confirm that it is in **Normal** mode, then securely insert the battery into the adapter, as shown in Figure 4.4a. Refer to Figure 4.4b for the charging setup.



(a) Connect the battery to the charging adapter



(b) Charger settings

Figure 4.4: Charging Setup for THEMIS Batteries

4.3.2 Power Delivery

TBC.

4.4 Computing Units

4.4.1 Main Computer

The main computer is a Beelink SER7 Mini PC with an 8-core AMD Ryzen 7 7840HS CPU (x86-64) at 5.1 GHz and a 12-core AMD Radeon 780M GPU at 2700 MHz. Detailed information can be found [here](#). It primarily handles THEMIS motion control.

4.4.2 Head Vision Computer

The head vision computer is using NVIDIA Jetson Orin NX 16GB (ONX 16GB), featuring an Ampere GPU + Arm Cortex-A78AE v8.2 64-bit CPU + 16 GB LPDDR5. Detailed information can be found [here](#). It is used as the main vision processor of THEMIS. A ZED 2i stereo camera is connected to the head computer as the main vision sensor.

4.4.3 Front Vision Computer

The front vision computer is using Jetson Orin NX 8GB (ONX 8GB) - Ampere GPU + Arm Cortex-A78AE v8.2 64-bit CPU + 8 GB LPDDR5. Detailed information can be found [here](#). The front vision computer is used for auxiliary vision processing of THEMIS. A ZED X Mini stereo camera is connected to the front computer as the auxiliary vision sensor.

5. Maintenance and Care

5.1 Regular Maintenance

Regular maintenance is critical for keeping your THEMIS Robot in optimal working condition. Here are some essential maintenance tasks:

1) Cleaning

- **Exterior Cleaning:** Use a soft, lint-free cloth to wipe down the outer surfaces of the robot. Avoid abrasive materials that could scratch the finish. For stubborn dirt, lightly dampen the cloth with water (without harsh chemicals) and gently clean.
- **Sensor and Camera Cleaning:** Use a microfiber cloth to carefully clean the lenses of the sensors and cameras. Dust or fingerprints on these lenses can affect the robot's perception and performance.
- **Internal Cleaning:** Typically, internal cleaning is not required. However, if the interior becomes dirty, please contact technical support for assistance.

2) Actuator and Motor Check

Regularly inspect the actuators and motors for signs of wear, unusual noises, or overheating. If any part seems faulty, contact technical support for advice or replacement options.

3) Joint Calibration

If the actuator's absolute-encoder data are lost or corrupted (e.g., limbs are asymmetrical after initialization even after several reboots), please follow the steps below to conduct zero-position calibration.

Step 1 - Adjust THEMIS to the calibration posture as shown in Figure 5.1. Use the calibration tools if necessary, as shown in Figure 5.2.

Step 2 - Log in to the main computer and run the calibration script. Follow the on-screen prompts to complete the calibration.

Terminal

```
1  cd THEMIS/THEMIS
2  python3 -m Util.bear_calibration
```



(a) Isometric view.



(b) Front view.

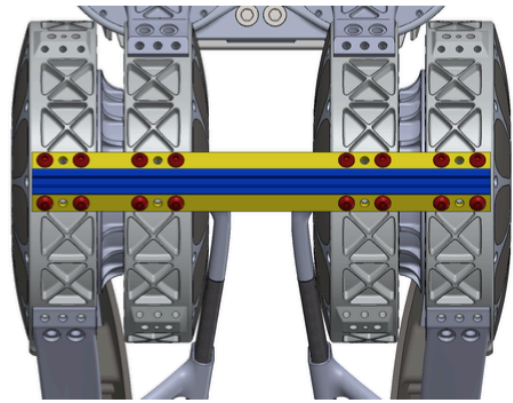


(c) Side view.

Figure 5.1: THEMIS Zero-Position Calibration Posture.



(a) Neck calibration tool.



(b) Hip calibration tool.

Figure 5.2: THEMIS Calibration Tools.

4) Firmware and Software Updates

Keep the robot's software and firmware up-to-date to ensure that it is running with the latest features, performance improvements, and security patches. Updates are released periodically, and instructions for updating can be found in the software manual or on our website.

5) Battery Maintenance

Regularly check the battery charge and health. Recharge the battery when its state of charge drops to approximately 20-30% to maintain its lifespan. Always use the recommended charger to avoid potential damage.

6) Mechanical Parts Check

Check the robot's joints and linkages for wear and tear. If any parts are loose or misaligned, tighten or realign them as necessary. Lubricate the moving parts lightly with an appropriate lubricant to reduce friction.

5.2 Storage Guidelines

Proper storage of your THEMIS Robot is crucial for protecting it when not in use. Follow these guidelines to ensure safe and effective storage:

1. **Power Off the Robot:** Before storing the robot, ensure it is powered off completely to prevent any accidental operation or battery drain.
2. **Store in a Dry, Temperature-Controlled Environment:** Store the robot in a dry and clean location, away from direct sunlight, extreme heat, or humidity. The recommended storage temperature range is between 0°C and 40°C (32°F to 104°F), and humidity should be kept below 85%. Avoid storing the robot in damp or humid environments, as moisture can damage internal components, especially electronic parts.

3. **Protect from Physical Damage:** For temporary storage, it is important to ensure the robot is securely positioned to prevent any risk of falling, sliding, or shifting. The robot can be placed on a stable surface such as a chair or suspended using a stable gantry crane. During storage, take precautions to protect the robot from dust and moisture. For long-term storage, we recommend returning the robot to the protective cases provided with the unit.
4. **Battery Storage:** If storing the robot for an extended period, ensure that the batteries remain within the recommended 40-60% storage charge range. This helps preserve battery life and prevents it from becoming over-discharged. Store the battery in a cool, dry place. Do not store the robot or battery in extreme temperatures to ensure safety and prevent degradation in robot and battery performance and service life.
5. **Disconnect Accessories:** If the robot has any connected accessories, such as cameras, sensors, or external devices, disconnect them before storage to prevent any damage. Store accessories separately in a safe place.
6. **Regular Storage Inspections:** If the robot is stored for a long period, perform periodic checks (e.g., every 1-2 months) to ensure that the battery is charged, there are no signs of corrosion, and the robot is still in good condition. Power on the robot occasionally to verify that the system remains operational.

By following these maintenance and storage guidelines, you can help ensure that your THEMIS robot operates reliably and stays in great condition for years to come. If you need assistance or have any questions, please don't hesitate to contact our support team.

6. Warranty and Support

6.1 Warranty Information

The THEMIS robot comes with a comprehensive standard warranty to ensure customer satisfaction and reliable performance. Below is a description of the warranty terms and coverage:

(a) Warranty Period

The standard warranty is valid for 12 months from the earlier of: (1) the date on which installation and customer acceptance are completed, or (2) the date of delivery.

(b) Warranty Coverage

This warranty covers the repair or replacement of parts that fail because of manufacturing defects under normal operating conditions.

- **Hardware:** The warranty covers malfunctions or failures of the robot's hardware components, including motors, sensors, joints, and structural elements, provided that the failure is not caused by misuse, physical damage, or operation outside the specified environmental conditions.
- **Software:** The warranty also covers software-related issues, such as bugs or errors in the operating system and robot control software. Issues caused by third-party software or incompatibility are excluded.

(c) Exclusions

- **Physical Damage:** Damage resulting from accidents, misuse, improper handling, or negligence is not covered.
- **Unauthorized Modifications:** The warranty is void if the robot has been modified or tampered with by unauthorized third parties.
- **Consumables:** Consumable components, including batteries, cables, and other consumable items, are not covered under this warranty.
- **Force Majeure:** Damage caused by external factors such as natural disasters, power surges, or other uncontrollable events is excluded.

(d) Warranty Transfer

The warranty is not transferable.

6.2 Support Contact Details

For any inquiries or assistance with your THEMIS robot under the standard warranty or any applicable active service plan, please feel free to contact us via the methods listed below:

- **Customer Support Email:** info@westwoodrobotics.io
- **Service Hours:** Monday to Friday, 9:00 AM to 5:00 PM (PST), except national holidays
- **Online Support Portal:**

Browse our FAQs at <https://www.westwoodrobotics.io/support/>

Join our Slack workspace for a better customer support experience:

https://join.slack.com/t/westwoodrobotics-cs/shared_invite/zt-25dhqdw7k-o4mepZtrxwzstCtZeitLhg

6.3 Service and Repair Information

(a) Warrant Service

- **Response Time:** Our customer support team will respond to warranty claims within within 24 business hours of receiving a service request. Initial troubleshooting will be conducted remotely to help resolve minor issues.

- Repairs and Replacements: If the issue cannot be resolved remotely, customers can send the robot or the affected component back to our service center, as instructed by technical support. We will assess the issue and provide the necessary solution at no cost, subject to the warranty terms.

(b) Shipping for Repairs

- Shipping for Service: If the robot or any component requires repair or replacement, customers are responsible for shipping the robot or the affected parts to our service center. We will cover the cost of shipping the repaired robot or repaired/replacement component back to the customer.

(c) Post-Warranty Support

- Continued Support: After the warranty period expires, we offer ongoing support through our maintenance and repair services at competitive rates. Customers may also opt for extended service packages for additional coverage beyond the warranty period.

The THEMIS standard warranty ensures that your robot remains in optimal condition for the first year of use, with full support for any manufacturing defects. If repairs or replacements are necessary, our team is ready to assist, ensuring minimal downtime and smooth operation.